Chapter 9: RL Circuit - Current Evolution

In the previous section, we established the potential difference across a coil:

$$u_{AB} = r \cdot i + L\frac{di}{dt}$$

When this coil is connected to a power source and a resistor, the circuit is governed by **Differential Equations**.

1. Establishment of Current (Growth Phase)

Consider a circuit with a generator (E), a resistor (R), and a coil (L, r) in series. Total resistance $R_t = R + r$.

The Differential Equation

Applying Kirchhoff's Loop Law ($E = u_{resistor} + u_{coil}$):

$$E = R \cdot i + (r \cdot i + L\frac{di}{dt}) = R_t \cdot i + L\frac{di}{dt}$$

Normalized to i:

$$\frac{L}{R_t}\frac{di}{dt} + i = \frac{E}{R_t}$$

The Solution: General Form & Derivation

The solution to this first-order linear differential equation is of the general form:

$$i(t) = A + Be^{-t/k}$$

1. Finding A (Steady State): As $t \to \infty$, the term $e^{-t/k} \to 0$. Therefore, the current $i(t) \to A$. This is the permanent phase where the current is maximum (I_{max}). Thus:

$$A = I_{max} = \frac{E}{R_t}$$

2. Finding k (Verification): Since $i(t)$ must verify the differential equation for all t, we substitute the general form and its derivative $\frac{di}{dt} = -\frac{B}{k}e^{-t/k}$ into the equation:

$$\frac{L}{R_t}\left(-\frac{B}{k}e^{-t/k}\right) + \left(I_{max} + Be^{-t/k}\right) = \frac{E}{R_t}$$

Since $I_{max} = \frac{E}{R_t}$, the constant terms cancel, leaving:

$$Be^{-t/k}\left(1 - \frac{L}{R_t \cdot k}\right) = 0$$

For this to be true for all t, we must have $1 - \frac{L}{R_t \cdot k} = 0$, which gives $k = \frac{L}{R_t}$. We define this as the **Time Constant** τ.

3. Finding B (Boundary Conditions): At $t = 0$, $i(0) = 0 \implies I_{max} + B = 0 \implies B = -I_{max}$.

Final Expression

$$i(t) = I_{max}\left(1 - e^{-t/\tau}\right)$$

2. Decay of Current (Discharge Phase)

When the source E is removed and the circuit is shorted ($E = 0$):

$$\frac{L}{R_t}\frac{di}{dt} + i = 0$$

Solution and Derivation

We use the general form $i(t) = Ce^{-t/k}$. Substituting i and $\frac{di}{dt} = -\frac{C}{k}e^{-t/k}$ into the ODE:

$$\frac{L}{R_t}\left(-\frac{C}{k}e^{-t/k}\right) + Ce^{-t/k} = 0 \implies Ce^{-t/k}\left(1 - \frac{L}{R_t \cdot k}\right) = 0$$

Again, we find $k = \tau = \frac{L}{R_t}$. At $t = 0$, the current starts at its initial value I_0 (often the I_{max} from a previous growth phase):

$$i(0) = I_0 \implies C = I_0$$

Final Expression

$$i(t) = I_0 \cdot e^{-t/\tau}$$

3. Graphical Interpretation

- **The Tangent Method:** The tangent to the growth curve $i(t)$ at $t = 0$ intersects the asymptote I_{max} at $t = \tau$. For decay, the tangent at $t = 0$ intersects the time axis at $t = \tau$.
- **Initial Slope:** At $t = 0$ for growth, $\left(\frac{di}{dt}\right)_0 = \frac{E}{L}$.